

Individual Weekly Report

CIXIO-Internship G3

Name: Ameen Anwar

Role: DevOps

1. Activities Completed

During the internship period, I focused primarily on DevOps, backend infrastructure, Linux administration, version control. My work involved both hands-on implementation and understanding how different modules integrate within the SmartHub ecosystem as well overall understanding of the different DevOps based tech-stacks.

Linux & System Administration

- Practiced extensive Linux command-line operations including:
 - File and directory management
 - User and permission management
 - Package installation and maintenance
 - Process monitoring and management
 - Service management using systemctl
 - Log inspection and troubleshooting
 - Networking Fundamentals

Pending Work: To explore more linux commands and operations and to be smooth while operating in linux.

Git & GitHub Version Control

- Learned Git fundamentals required for collaborative software development.
- Practiced:
 - Repository cloning
 - Branch creation and management
 - Staging and committing changes

- Pulling and pushing updates
- Merging branches
- Resolving merge conflicts
- Understood Pull Request workflows and collaborative development practices.
- Learned the purpose and usage of:
 - .gitignore
 - GitHub Issues
 - GitHub Projects
 - Repository management

Pending Work: Various git commands yet to explore.

Docker & Containerization

- Learned the fundamentals of Docker.
- Understood:
 - Images and Containers
 - Dockerfiles
 - Container lifecycle
 - Volume mounting
- Practiced running applications inside containers.
- Learned how Docker simplifies deployment consistency across environments.

Pending: To learn Docker Compose and further exploration of Docker.

Backend Development & Authentication

- Reviewed backend web architecture fundamentals.
- Built and tested Express.js applications.
- Learned authentication concepts including:
 - Authentication vs Authorization
 - Password hashing with bcrypt
 - Salt generation
 - User registration workflows
 - Login verification workflows

- JWT (JSON Web Tokens)
- Access Tokens
- Refresh Tokens
- Protected Routes
- Basic OAuth concepts

JavaScript & Node.js Development

- Strengthened JavaScript programming fundamentals.
- Worked with:
 - Functions and callbacks
 - Closures
 - Arrow functions
 - Array methods
 - ES6+ features
 - Modules and imports/exports
- Learned Node.js core modules:
 - fs
 - path
 - os
- Developed small backend applications using Express.js.

Pending Work: Need to refresh JS, and the MERN stack.

Mobile Application Development

- Set up Flutter development environment on Linux.
- Learned:
 - Widget architecture
 - Stateful and Stateless widgets
 - Application lifecycle
 - Local storage mechanisms
 - Offline-first concepts
- Built a small Flutter task tracking application for practical learning.

Pending Work: To learn the dart language and detailed understanding of mobile development.

AI & Local LLM Deployment

- Installed and configured Ollama for local AI model execution.
- Learned how local Large Language Models can be hosted and served.
- Studied FastAPI fundamentals.
- Ran local AI services and experimented with self-hosted AI deployment workflows.
- Understood the relationship between AI models, APIs, and client applications.

Pending Work: Need to explore other AI models for hosting other ollama. Need to be familiarized with concepts of python for studying LLM and AIs.

2. System Architecture & DevOps Understanding

Through internship discussions, self-learning, and project planning, I developed an understanding of how large-scale software systems are designed and deployed.

SmartHub Architecture

- Understood how frontend, backend, mobile, AI, and infrastructure modules interact.
- Learned service-based architecture concepts.
- Studied API-driven communication between modules.
- Understood the importance of scalability and maintainability in modern applications.

DevOps Concepts Learned

- Continuous Integration (CI)
- Continuous Deployment (CD)
- Infrastructure automation
- Environment configuration management
- Deployment pipelines
- Monitoring and observability

Future Focus Areas

Moving forward, I plan to deepen my knowledge in:

- Docker and Docker Compose
 - Kubernetes
 - CI/CD Pipelines
 - GitHub Actions
 - Cloud Infrastructure
 - Monitoring & Observability
 - Infrastructure as Code
 - Advanced Backend Development
 - Production Deployment Strategie
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3. Professional Growth & Key Takeaways

Technical Growth

- Improved confidence working in Linux-based development environments.
- Developed a stronger understanding of backend and infrastructure technologies.
- Gained practical exposure to modern DevOps tools and workflows.
- Learned how deployment and operations complement software development.

Problem Solving

- Learned to troubleshoot installation, networking, and deployment issues independently.
- Improved ability to learn unfamiliar technologies quickly.

Team & Project Awareness

- Understood how different technical domains contribute to a complete software product.
- Learned the importance of documentation and version control.
- Recognized the value of maintainable architecture and automation.
- Succeeded in conducting proper communication with team members.

Summary

This internship provided exposure to multiple areas of modern software engineering, particularly DevOps, backend systems, Linux administration, containerization, authentication systems, mobile development fundamentals, and AI deployment technologies. The experience strengthened both technical foundations and professional problem-solving skills while preparing me for contributing to larger production-grade software systems. Looking forward to strengthen knowledge mainly in DevOps and also other domains as well.

Github profile :<https://github.com/ameenanwar>